

CHIRANJEEV KUMAR

Ludhiana, Punjab, India | +91 7527906046 | chiranjeevk171@gmail.com

[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

EDUCATION

Gulzar Group of Institutions — B.Tech in CSE (Specialization: AI/ML), Affiliated to IKGPTU, Jalandhar

Expected Graduation: June 2028

Current CGPA: 7.41 / 10

Relevant Coursework: Data Structures & Algorithms, OOPs, DBMS, Operating Systems, Computer Networks (CN)

TECHNICAL SKILLS

Languages: C++, JavaScript (ES6+), Python, HTML5, CSS3, SQL

Frameworks & Libraries: React.js, Node.js, Express.js, Tailwind CSS, Standard Template Library (STL)

Databases & Developer Tools: MongoDB, Git, GitHub

TECHNICAL PROJECTS

Personal Productivity Progressive Web App (PWA) — [Live Demo](#) [GitHub](#)

Full-Stack Developer

- Engineered an offline-first productivity workspace that achieved a 40% reduction in perceived task-tracking latency compared to standard blocking network architectures by implementing Optimistic UI updates and background asset compilation
- Implemented an asynchronous outbox synchronization queue using custom Service Workers and local storage snapshots, guaranteeing 100% data preservation and automatic background replay upon network reconnection
- Designed a component-based React architecture with localized state re-validation hooks, enabling modular, reusable UI elements across core task execution, fluid goal tracking, and live-computed calendar modules
- Built secure, stateless RESTful APIs utilizing Node.js, Express.js, and JWT authentication guards, integrated with a cloud-hosted MongoDB cluster for cross-session persistent storage
- Technologies used:** React.js, Node.js, Express.js, MongoDB, Service Workers, Workbox, JWT, Git

WebAssembly Huffman File Compressor — [Live Demo](#) [GitHub](#)

Full-Stack Systems Developer

- Built a browser-based, privacy-first file compression tool by compiling a custom C++17 Huffman Encoding engine to WebAssembly via Emscripten, achieving near-native execution speed for client-side compression
- Engineered a cross-language memory bridge using Emscripten's ccall and the MEMFS virtual filesystem to safely transfer raw binary data between JavaScript and the native C++ engine without memory leaks
- Designed a responsive frontend using the FileReader and Blob APIs to process files entirely client-side — no uploads, no server processing, ensuring user data never leaves the device
- Automated cross-compilation and deployment via a GitHub Actions CI/CD pipeline, building optimized .wasm/.js binaries and continuously deploying to Vercel
- Achieved a 49% compression ratio on text data (114 KB → 57.86 KB) using a min-heap-based Huffman tree and bitset-driven bit-packing, validated via byte-for-byte lossless decompression
- Technologies used:** C++17, WebAssembly, Emscripten, JavaScript (ES6), HTML5/CSS3, GitHub Actions, Vercel, Git

EXPERIENCE & TRAINING

Software Engineer Job Simulation — Skyscanner (Forage) [Certificate](#)

June 2026 – July 2026

Software Engineer | Ludhiana, Punjab

- Reconstructed the application architecture to ensure clean separation of components and robust state-handling in React
- Developed a date picker component utilizing Skyscanner's custom Backpack library
- Wrote and executed automated tests using Jest to verify correct application rendering and behavior
- Learned how to debug and optimize build configurations, including package compilation and asset packaging, on native developer environments

ACHIEVEMENTS & CERTIFICATIONS

- Smart India Hackathon (SIH) Internal Hackathon:** Secured **2nd Position** among competing institutional teams by designing and developing an innovative software solution under a competitive timeline. [\[Certificate\]](#)
- Cloud Consultant Certification** — PMKVY (NSQF Level 6), IT-ITeS Sector Skill Council, NASSCOM-recognized; 600 hrs, Grade A. [\[Certificate\]](#)
- Digital Logo Design:** Secured 2nd position in Digital Logo Designing at GNE'S ACME 2025, an inter-college competition organized by Guru Nanak Dev Engineering College. [\[Certificate\]](#)